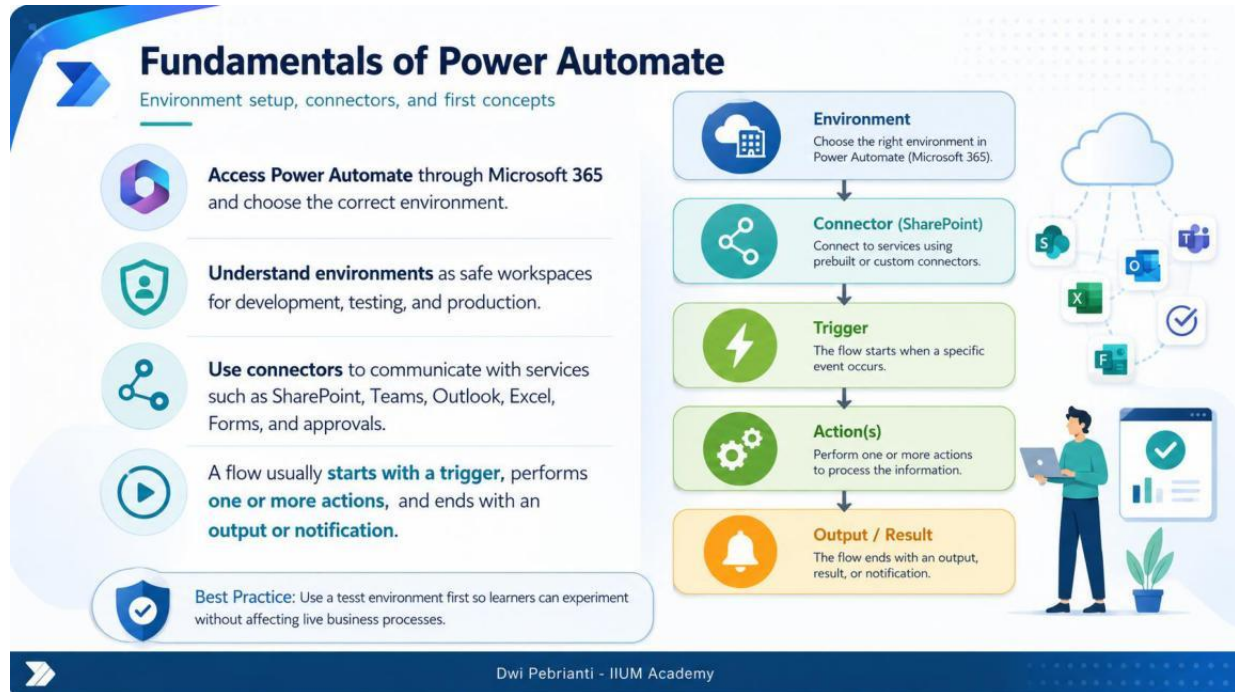


POWER AUTOMATE

Student Laboratory Manual

Zero to Hero for Insurance Professionals



Prepared for:

IIUM Academy / Prudential BSN Takaful Training

Prepared by:

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27 July 2026

Manual objective

This manual converts the training slides into executable laboratory activities. Every build starts with selecting the correct flow type and ends with a validation step using Flow Checker, test execution, run history, and an expected result.

How to Use This Manual

This manual is based on the 58-slide Power Automate training deck. The labs deliberately reuse the same insurance examples so that learners move from simple automation to a complete claims approval workflow.

Learning Outcomes

- Choose Automated, Instant, or Scheduled cloud flows for the correct business trigger.
- Connect SharePoint, Outlook, OneDrive, RSS, and Approvals securely.
- Use triggers, actions, Dynamic Content, conditions, expressions, and variables.
- Build branded HTML claim reports.
- Use Copilot to draft a cloud flow from natural language and then validate it.
- Build a complete claims approval and audit-trail workflow.
- Use Flow Checker, manual tests, run history, and edge-case tests before considering a flow complete.

Lab Map

Lab	Activity	Suggested time
0	Pre-lab setup: environment & connections	10-15 min
1	Scheduled daily reminder	15 min
2	Automatically save email attachments	20 min
3	Risk-keyword monitoring with RSS	20 min (optional)
4	New claim acknowledgement with Dynamic Content	20 min
5	Conditional routing by claim type	20 min
6	Expressions & variables for claim data	25-30 min
7	HTML/CSS claim summary report	30 min
8	Copilot: Describe it to design it	20-25 min
9	Capstone: claims approval, routing & audit log	45-60 min
10	Final validation, troubleshooting & submission	10 min

Pre-lab Requirements

- Microsoft 365 account with Power Automate access.
- SharePoint site where you are allowed to create lists.
- Outlook mailbox under the same Microsoft 365 account.
- OneDrive for Business or a SharePoint document library for attachment storage.
- For approvals: a valid internal Microsoft 365 approver account and an environment where Approvals/Dataverse is available.
- Use a test or developer environment whenever one is provided. Avoid production customer data.

Currency convention

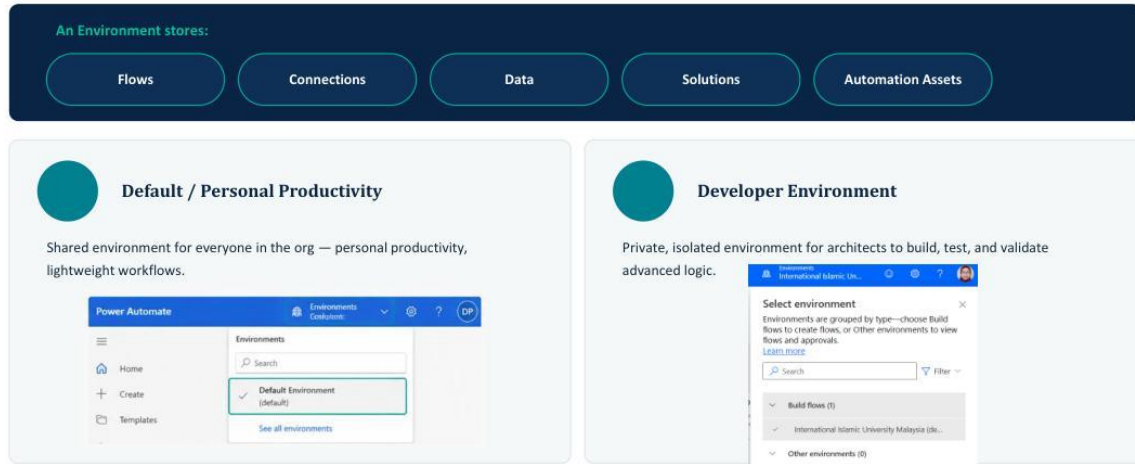
General expression/Copilot examples use US dollars (\$) to match the training slides. The Takaful capstone uses RM because the capstone slides explicitly use RM.

Lab 0 - Pre-Lab Setup: Environment and Connections

MODULE 2 · ECOSYSTEM

The Concept of Environments

An Environment is more than a folder — it's a storage container and a security boundary for automation assets.



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Reference slide: environment selection and separation of Default vs Developer environments.

Goal: confirm that you can access Power Automate, select the correct environment, and authenticate the services used in later labs.

Step 1. Open Power Automate.

Open a browser and go to <https://make.powerautomate.com>. Sign in with your training Microsoft 365 account.

Step 2. Select the training environment.

- Use the Environment selector in the top-right area.
- Choose the designated Developer/Test/Training environment if provided.
- If no dedicated environment exists, use the environment specified by the instructor.

Step 3. Open Connections.

- From the left navigation, open Connections (sometimes under More).
- Confirm that SharePoint and Office 365 Outlook are connected.
- For later labs, also confirm OneDrive for Business and Standard approvals if available.

Step 4. Repair a connection if needed.

- If a connection shows an error, select it and sign in again.
- Use the same training identity unless your instructor specifies a service account.
- Do not enter another participant's credentials.

Validation

Check	Pass condition	If not passed
Environment	Correct training environment is shown in the header.	Switch environment before creating any flow.
SharePoint connection	Connection is available without warning.	Re-authenticate or ask the instructor for site permission.
Outlook connection	Connection is available without warning.	Re-authenticate and verify mailbox access.

PASS criterion

Do not continue until Power Automate opens in the intended environment and both SharePoint and Outlook connections are usable.

Lab 1 - Scheduled Daily Reminder

Slide task: “Daily Reminders - Scheduled recurrence flows send custom alerts via Outlook or Gmail.” This lab uses Outlook.

Scenario

Every morning, an insurance officer should receive a reminder to review policy renewals that need follow-up.

Step 1. Select the correct flow type.

- Power Automate > Create.
- Select Scheduled cloud flow.
- Flow name: Claims - Daily Renewal Reminder.
- For classroom testing, set Repeat every = 1 minute. After the test, change it to daily.

Step 2. Configure the Recurrence trigger.

- Confirm the recurrence frequency.
- If a Time zone option is shown, select your local training time zone.
- For the final version, use a daily schedule at the business-approved time (for example 8:00 AM).

Step 3. Add the Outlook action.

- Select + > Add an action.
- Search for Office 365 Outlook.
- Choose Send an email (V2).
- Rename the action: Send Daily Renewal Reminder.

Step 4. Configure the reminder email.

- To: your own training email address.
- Subject: Daily Reminder - Policy Renewals Requiring Follow-up.
- Body: Good morning. Please review policies approaching renewal and complete the required customer follow-up.

Step 5. Save and check the flow.

- Select Save.
- Run Flow Checker and resolve any errors.

Step 6. Test.

- Wait for the temporary one-minute recurrence.
- Open My flows > the flow > Run history.
- Open the latest run and confirm Recurrence and Send Daily Renewal Reminder both have green checks.
- Confirm the reminder email arrives in Outlook.

Step 7. Return the schedule to the intended production training setting.

- Change the recurrence from 1 minute to the intended daily schedule.
- Save again.

PASS criterion

A successful run appears in Run history and the expected reminder email is received exactly once for the test run.

Lab 2 - Automatically Save Email Attachments

Slide task: automatically download policy documents or claim photos to a cloud folder.

Prepare the destination

- In OneDrive for Business, create a folder named Claims_Attachments.
- Keep a small test file ready, such as test-claim.txt or a small PDF.

Step 1. Create an Automated cloud flow.

- Power Automate > Create > Automated cloud flow.
- Name: Claims - Save Email Attachments.
- Trigger: Office 365 Outlook - When a new email arrives (V3).

Step 2. Configure the email trigger.

- Folder: Inbox.
- Only with Attachments: Yes.
- Subject Filter: LAB ATTACHMENT (recommended for safe testing).
- For reliable attachment handling, leave Include Attachments = No and retrieve the attachment explicitly in the next steps.

Step 3. Add Apply to each.

- Add action > Control > Apply to each.
- For Select an output from previous steps, choose Attachments from the Outlook trigger.

Step 4. Inside the loop, add Get Attachment (V2).

- Connector: Office 365 Outlook.
- Message Id: select Message Id from the trigger.
- Attachment Id: select Attachment Id from the current attachment item.

Step 5. Inside the loop, add OneDrive for Business - Create file.

- Folder Path: /Claims_Attachments.
- File Name: choose the attachment Name.
- File Content: choose Content Bytes from Get Attachment (V2).
- Rename the action: Save Attachment to OneDrive.

Step 6. Save and run Flow Checker.

Step 7. Test the flow.

- Send an email to your training mailbox with subject LAB ATTACHMENT and attach a small file.
- Open the flow Run history and confirm all actions are green.
- Open OneDrive > Claims_Attachments and confirm the file exists and opens correctly.

PASS criterion

The test attachment appears in the correct OneDrive folder with the original filename and valid file content.

Troubleshooting

If the flow does not trigger, check the Subject Filter and Only with Attachments setting. If the file is created but cannot open, verify that File Content comes from Get Attachment (V2), not from the attachment metadata.

Lab 3 - Risk-Keyword Monitoring with RSS (Optional)

Slide task: automated monitoring of industry keywords for market intelligence. RSS is used because it provides a standard event source without requiring a premium web-search connector.

Instructor note

RSS triggers poll periodically and may not fire immediately. Build the deployed version as an Automated cloud flow, but validate the logic with a known active RSS feed or the instructor's prepared feed.

Step 1. Create a SharePoint destination list.

- Create a blank list named Market_Intelligence.
- Use Title for Headline.
- Add SourceLink (Single line of text), PublishedDate (Date and time), and Keyword (Single line of text).

Step 2. Create an Automated cloud flow.

- Create > Automated cloud flow.
- Name: Risk Intelligence - RSS Keyword Monitor.
- Trigger: RSS - When a feed item is published.

Step 3. Configure the RSS trigger.

- RSS feed URL: use a valid public feed supplied by the instructor.
- Chosen property: PublishedOn/Published date if available.

Step 4. Add a Condition.

- Left value: Feed title.
- Operator: contains.
- Right value: catastrophic (or another training keyword).
- Rename: Does headline contain risk keyword?

Step 5. In the True branch, add SharePoint - Create item.

- List: Market_Intelligence.
- Title: Feed title.
- SourceLink: Primary feed link.
- PublishedDate: Feed published on.
- Keyword: catastrophic.

Step 6. Save, Flow Checker, and test.

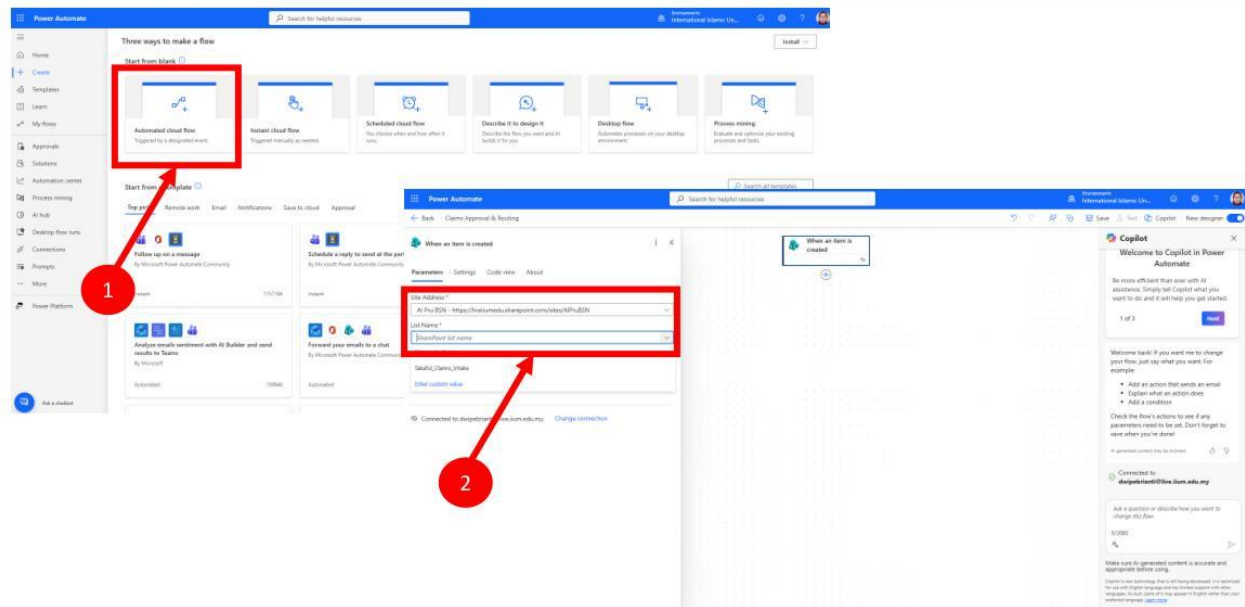
- If a new matching item is published, verify a SharePoint row is created.
- If the instructor provides a test feed, publish a matching item and check Run history.
- Confirm the False branch does not create a row.

PASS criterion

A matching feed item creates one SharePoint record containing the headline, link, date, and keyword.

Lab 4 - New Claim Acknowledgement with Dynamic Content

Create the Flow and Set The Trigger



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Reference slide: selecting Automated cloud flow and configuring the SharePoint trigger.

Part A - Create the SharePoint list

Create a blank SharePoint list named Takaful_Claims_Intake. To keep the lab runnable, this manual adds ClaimantEmail to the fields shown in the slides.

Column	Type	Recommended setting
Title	Single line of text	Use as ClaimantName; rename the visible column to ClaimantName.
ClaimantEmail	Single line of text	Example: your training email address.
CertificateNo	Single line of text	Example: CERT-1001.
ClaimType	Choice	Motor; Property; Health; Other.
IncidentDate	Date and time	Date only is sufficient.
ClaimAmount	Number	2 decimal places.
Description	Multiple lines of text	Short incident description.
Status	Choice	New; Under Review; Approved; Rejected.

SharePoint naming note

Renaming Title to ClaimantName changes the visible name, but SharePoint may retain the internal name Title. For beginner labs, select the ClaimantName token from Dynamic Content rather than typing the internal name manually.

Part B - Build the flow

Step 1. Create an Automated cloud flow.

- Name: Claims - New Item - Send Acknowledgement.
- Trigger: SharePoint - When an item is created.

Step 2. Configure and rename the trigger.

- Site Address: your training SharePoint site.
- List Name: Takaful_Claims_Intake.
- Rename the trigger: When a new claim is created.

Step 3. Add Outlook - Send an email (V2).

- Rename: Email claimant - acknowledgement.
- To: ClaimantEmail from Dynamic Content.
- Subject: Claim received - [CertificateNo]. Insert CertificateNo from Dynamic Content.

Suggested body:

Dear [ClaimantName],

We have received your claim.

Certificate No: [CertificateNo]

Claim Type: [ClaimType]

Incident Date: [IncidentDate]

Claim Amount: [ClaimAmount]

Your claim is now under review.

Regards,

Claims Team

Replace each bracketed item with the corresponding Dynamic Content token.

Step 4. Save, Flow Checker, and test.

- Add a new SharePoint item using your own email as ClaimantEmail.
- Confirm the flow run is successful.
- Confirm the email contains the exact values entered in SharePoint and no manual retyping is required.

PASS criterion

Creating a new SharePoint claim automatically sends one personalized acknowledgement to the email stored in ClaimantEmail.

Lab 5 - Conditional Routing by Claim Type

This lab implements the slide example: IF Claim Type = Motor, route to the Motor Claims Department; otherwise route to General Claims.

Step 1. Create an Automated cloud flow.

- Name: Claims - Route by Type.
- Trigger: SharePoint - When an item is created.
- Site/List: your Takaful_Claims_Intake list.

Step 2. Add Control - Condition.

- Rename: Is this a motor claim?
- Left: ClaimType Value from Dynamic Content.
- Operator: is equal to.
- Right: Motor.

Step 3. Configure the True branch.

- Add Outlook - Send an email (V2).

- For classroom testing, send to yourself.
- Subject: MOTOR CLAIM - [CertificateNo].
- Body: This claim must be routed to the Motor Claims Department.

Step 4. Configure the False branch.

- Add Outlook - Send an email (V2).
- For classroom testing, send to yourself.
- Subject: GENERAL CLAIM - [CertificateNo].
- Body: This claim must be routed to the General Claims Department.

Step 5. Save and run Flow Checker.

Step 6. Test both branches.

- Test A: create a claim with ClaimType = Motor. Confirm the True branch runs and the MOTOR CLAIM email arrives.
- Test B: create a claim with ClaimType = Property. Confirm the False branch runs and the GENERAL CLAIM email arrives.
- Open each run in Run history and verify the unused branch is shown as skipped.

PASS criterion

Both True and False test cases route correctly, and exactly one branch executes for each claim.

Lab 6 - Expressions and Variables for Claim Data

MODULE 2 • TOOLKIT — MUST KNOW

Expressions: Formatting & Calculating Data

Expressions format, manipulate, or calculate data — dates, currency, text, and math — inside a flow step.

Date Formatting

```
formatDateTime(triggerOutputs()?['body/Date'], 'dd-MM-yyyy')
```

Example output: **15-Jan-2026**

Currency Formatting

```
formatNumber(variables('ClaimAmount'), 'C2')
```

Example output: **\$5,000.00**

Other everyday expressions

- Extracting text (e.g. pulling a policy number from a string)
- Performing mathematical calculations (e.g. premium totals)
- Combining strings (e.g. building a full claim reference)

Function	What it does	Example
formatDateTime()	Formats date/time	15-Jan-2026
formatNumber()	Formats numbers/currency	\$5,000.00
add()	Performs calculation	5000 + 125 = 5125
concat()	Combines text values	POL45821-20260115

Reference slide: formatDateTime(), formatNumber(), add(), and concat().

Part A - Create expression-practice data

Create a separate SharePoint list named Insurance_Claims_Expression with the following columns:

Field	Type	Test value
Title	Single line of text	Rename visible column to Claimant; example Ahmad.

ClaimantEmail	Single line of text	Use your test email.
ClaimDate	Date and time	15-Jan-2026 09:30.
ClaimAmount	Number	5000.
AdminFee	Number	125.
PolicyNumber	Single line of text	POL45821.

Part B - Build the expression flow

Step 1. Create an Automated cloud flow.

- Name: Claims - Format and Calculate Data.
- Trigger: SharePoint - When an item is created.
- List: Insurance_Claims_Expression.

Step 2. Add Compose - Format Claim Date.

Use the fx Expression editor, not ordinary text. Enter:

```
formatDateTime(triggerBody()?['ClaimDate'],'dd-MMM-yyyy')
```

Expected output: 15-Jan-2026.

Step 3. Add Compose - Format Claim Amount.

```
formatNumber(triggerBody()?['ClaimAmount'],'C2','en-US')
```

Expected output: \$5,000.00.

Step 4. Add Compose - Calculate Total.

```
add(triggerBody()?['ClaimAmount'],triggerBody()?['AdminFee'])
```

Expected output: 5125.

Step 5. Add Compose - Format Total.

```
formatNumber(add(triggerBody()?['ClaimAmount'],triggerBody()?['AdminFee']),'C2','en-US')
```

Expected output: \$5,125.00.

Step 6. Add Compose - Build Claim Reference.

```
concat(triggerBody()?['PolicyNumber'],'-',formatDateTime(triggerBody()?['ClaimDate'],'yyyyMMdd'))
```

Expected output: POL45821-20260115.

Part C - Add variables

Step 7. Immediately after the trigger, add Initialize variable - TotalAmount.

- Name: TotalAmount.
- Type: Float.
- Value: 0.

Step 8. Add Initialize variable - ClaimReference.

- Name: ClaimReference.
- Type: String.
- Value: leave blank or use an empty string.

Step 9. After initialization, add Set variable - TotalAmount.

```
add(triggerBody()?['ClaimAmount'],triggerBody()?['AdminFee'])
```

Step 10. Add Set variable - ClaimReference.

```
concat(triggerBody()?['PolicyNumber'],'-',formatDateTime(triggerBody()?['ClaimDate'],'yyyyMMdd'))
```

Variable rule

Initialize variables at the top level of the flow, not inside a Condition, Scope, or Apply to each. They are global within the flow.

Step 11. Save and test.

- Create a test item with the values in the table above.
- Open the run and inspect each Compose action's Outputs.
- Inspect the Set variable actions and confirm TotalAmount = 5125 and ClaimReference = POL45821-20260115.
- If the output displays the text formatDateTimes(...) or concat(...), the formula was typed as plain text. Delete it and re-enter it through fx > Expression.

PASS criterion

All five expression outputs and both variables match the expected values.

Lab 7 - HTML/CSS Claim Summary Report

MODULE 2 • TOOLKIT — MUST KNOW

Advanced Interface & AI-Assisted Design

Describe It to Design It

Power Automate can use AI-assisted design to draft a workflow from a natural-language description you type in plain English.

```

graph TD
    A[Manually trigger a flow] --> B[Get items]
    B --> C[Select]
    C --> D[Create HTML table]
    D --> E[Style HTML Table]
    E --> F[Final Claims Report]
    F --> G[Send an email (V2)]
            
```

Insurance Claim Summary Report Summarize this email AI PRU INC

DWI PEBRIANTI To: @ DWI PEBRIANTI Reply Reply all

Claimant	Claimant Email	Claim Date	Claim Amount	Admin Fee	Policy Number
Ahmad	dwi@pebrianti@live.iium.edu.my	14-Jan-2026	\$5,000.00	\$125.00	POL45821
Dwi	dwi@pebrianti@live.iium.edu.my	02-Jul-2026	\$10,000.00	\$500.00	POL12345
Uy chan	dwi@pebrianti@live.iium.edu.my	13-Jul-2026	\$12,000.00	\$1,200.00	POL43432
Wawa	dwi@pebrianti@live.iium.edu.my	11-Jul-2026	\$13,000.00	\$1,500.00	POL98765
Luhur	dwi@pebrianti@live.iium.edu.my	27-Apr-2026	\$4,500.00	\$130.00	POL22556

AI PRU INSURANCE Claim Summary Report Generated on 26-Jul-2026

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Reference slide: final claim-summary email and the flow used to build it.

Goal: retrieve claim rows from SharePoint, keep only required columns, format them, build an HTML table, style it, and email the report.

Step 1. Create an Instant cloud flow.

- Create > Instant cloud flow.
- Name: Claims - Generate HTML Summary.
- Trigger: Manually trigger a flow.

Step 2. Add SharePoint - Get items.

- Rename: Get Claims.
- List: Insurance_Claims_Expression (or the instructor's designated claims list).

Step 3. Add Data Operations - Select.

- Rename: Select Report Columns.

- From: value from Get Claims.
- IMPORTANT: type the column labels on the LEFT and put Dynamic Content/expressions on the RIGHT.

LEFT: fixed label	RIGHT: value
Claimant	Title / Claimant dynamic content
Claimant Email	ClaimantEmail
Claim Date	fx: formatDateTime(item()?['ClaimDate'],'dd-MMM-yyyy')
Claim Amount	fx: formatNumber(item()?['ClaimAmount'],'C2','en-US')
Admin Fee	fx: formatNumber(item()?['AdminFee'],'C2','en-US')
Policy Number	PolicyNumber

Step 4. Add Data Operations - Create HTML table.

- Rename exactly: Create HTML table.
- From: Output from Select Report Columns.
- Columns: Automatic.

Step 5. Add Compose - Style HTML Table.

Rename the action exactly before adding the expression. Click the Inputs field > fx > Expression and paste:

```
replace(
  replace(
    replace(
      body('Create_HTML_table'),
      '<table>',
      '<table style="width:100%;border-collapse:collapse;font-family:Arial,sans-serif;">'
    ),
    '<th>',
    '<th style="background:#0795a5;color:white;padding:10px;text-align:left;border:1px solid #dddddd;">'
  ),
  '<td>',
  '<td style="padding:10px;border:1px solid #dddddd;">'
)
```

Step 6. Add Compose - Final Claims Report.

Click Inputs > fx > Expression and paste:

```
concat(
  '<div style="font-family:Arial,sans-serif;max-width:900px;border:1px solid #dddddd;">',
  '<div style="background:#0b2d55;color:white;padding:18px 24px;">',
  '<span style="font-size:22px;font-weight:bold;">AI PRU INSURANCE</span>',
  '<span style="float:right;text-align:right;font-size:13px;">Claim Summary Report<br>Generated on ',
  formatDateTime(utcNow(),'dd-MMM-yyyy'),
  '</span></div>',
  '<div style="padding:20px;">',
  outputs('Style_HTML_Table'),
  '</div></div>'
)
```

Step 7. Add Outlook - Send an email (V2).

- To: your own training email.
- Subject: Insurance Claim Summary Report.
- Body: select Outputs from Final Claims Report as Dynamic Content. Do not paste the concat() expression into the email body.

Step 8. Save, Flow Checker, and test.

- Run the Instant flow manually.
- Open Run history and inspect Style HTML Table: the output should start with <table style=, not replace(.
- Inspect Final Claims Report: the output should start with <div style=, not concat(.
- Open Outlook and confirm only the selected columns appear.

Common error 1

If the email literally shows replace(...) or concat(...), the formula was entered as ordinary text. Re-enter it through the fx Expression editor.

Common error 2

If the table headers contain dates or numbers, the Select mapping is reversed. Fixed labels belong on the LEFT; Dynamic Content belongs on the RIGHT.

PASS criterion

The email contains a navy report header, a teal table-header row, and only the intended SharePoint fields. No SharePoint system metadata or raw expressions are visible.

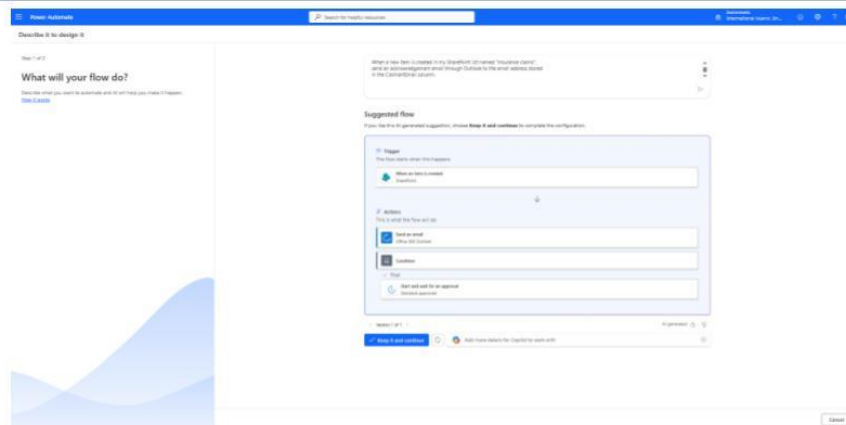
Lab 8 - Copilot: “Describe It to Design It”

MODULE 2 • TOOLKIT — MUST KNOW

Advanced Interface & AI-Assisted Design

Try typing this into Power Automate's AI builder:

“When a new insurance claim is added to SharePoint, email the claimant, and if the claim exceeds \$5,000, send it to a supervisor for approval.”



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Reference slide: natural-language prompt and Copilot-generated flow structure.

Terminology

For this task use Power Automate Copilot / “Describe it to design it.” AI Builder is a different Power Platform capability for AI models and AI actions.

Prompt from the training slides:

When a new insurance claim is added to SharePoint, email the claimant, and if the claim exceeds \$5,000, send it to a supervisor for approval.

Step 1. Open Power Automate Home and start Copilot flow creation.

- Use Create your automation with Copilot / Describe it to design it.
- Paste the prompt above and select Generate.

Step 2. Review the suggested structure.

- Expected trigger: SharePoint - When an item is created.
- Expected action: Send an email.
- Expected condition: ClaimAmount > 5000.
- Expected True branch: Start and wait for an approval.
- If the structure is reasonable, select Keep it and continue.

Step 3. Check all connections.

- SharePoint, Outlook, and Approvals should have valid connections.
- Resolve any red connection warning before creating the flow.

Step 4. Configure the SharePoint trigger.

- Site: training site.
- List: Insurance_Claims_Expression or the instructor's insurance list.

Step 5. Configure claimant acknowledgement.

- To: ClaimantEmail.
- Insert Claimant/Title, PolicyNumber, ClaimDate, and ClaimAmount using Dynamic Content.

Step 6. Configure the condition.

- ClaimAmount is greater than 5000.
- Use numeric 5000, not the text \$5,000.

Step 7. Configure Start and wait for an approval.

- Approval type: Approve/Reject - First to respond.
- Assigned to: a valid internal Microsoft 365 email address, preferably your own account for the first test.
- Title: High-Value Insurance Claim Approval - [PolicyNumber].
- Details: include claimant, policy number, claim date, and claim amount.

Step 8. Save and test two cases.

- Case A: ClaimAmount = 3000. Expected: acknowledgement email; approval branch skipped.
- Case B: ClaimAmount = 7500. Expected: acknowledgement email; approval request created.
- For Case B, respond Approve or Reject. Start and wait for an approval remains Running until a human responds; this is expected behavior.

PASS criterion

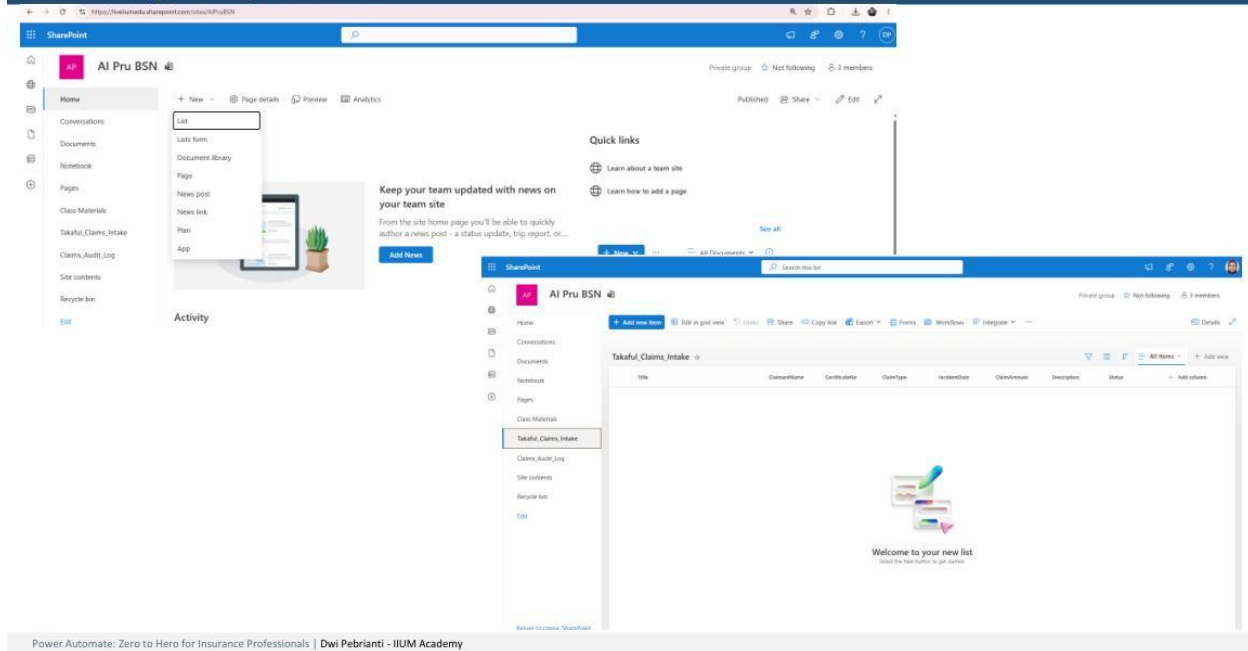
The low-value claim skips approval, while the high-value claim pauses for a valid approval request and continues after an approver responds.

Approval troubleshooting

If you see InvalidApprovalCreateRequestAssignedToNoValidUsers, replace the Assigned to value with a valid internal Microsoft 365 user. If the Approvals connector cannot load, reconnect it or ask the instructor to confirm Dataverse/Approvals availability in the environment.

Lab 9 - Capstone: Claims Approval, Routing and Audit Log

Creating Sharepoint Lists



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Reference slide: two-list architecture - claim intake and audit log.

This lab implements the reusable pattern from the slides: Trigger -> Notify -> Evaluate -> Route to human -> Record.

Part A - Create the two SharePoint lists

1. Takaful_Claims_Intake

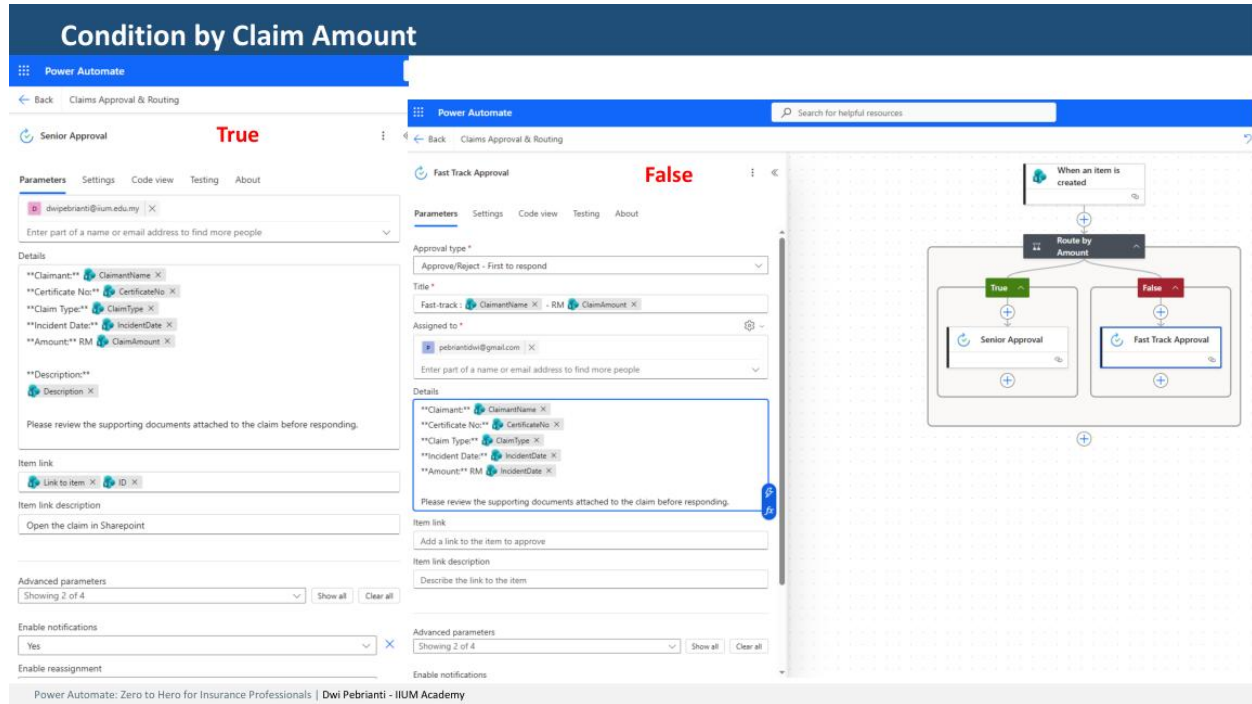
Column	Type	Notes
Title	Single line of text	Rename visible column to ClaimantName.
ClaimantEmail	Single line of text	Added so the workflow can email the claimant.
CertificateNo	Single line of text	Claim/certificate reference.
ClaimType	Choice	Motor, Property, Health, Other.
IncidentDate	Date and time	Date only.
ClaimAmount	Number	2 decimals; capstone examples use RM.
Description	Multiple lines of text	Incident details.
Status	Choice	New, Under Review, Approved, Rejected.

2. Claims_Audit_Log

Column	Type	Notes
Title	Single line of text	Rename visible column to ClaimID or use it for the source item ID.
ClaimantName	Single line of text	Copy from intake.
Approver	Single line of text	Approver email/name.

Outcome	Single line of text	Approved or Rejected.
Comments	Multiple lines of text	Approval comments.
DecisionTimestamp	Date and time	Use utcNow() or approval completion time.

Part B - Build the skeleton



Reference slide: amount-based branching to Senior Approval and Fast Track Approval.

Step 1. Create an Automated cloud flow.

- Name: Claims Approval & Routing.
- Trigger: SharePoint - When an item is created.
- Site/List: Takaful_Claims_Intake.
- Rename trigger: When a new claim is created.

Step 2. Add Outlook acknowledgement before the decision.

- Action: Send an email (V2).
- Rename: Email claimant - intake received.
- To: ClaimantEmail.
- Subject: We received your takaful claim - [CertificateNo].

Step 3. Add Condition - Route by Amount.

- Left: ClaimAmount.
- Operator: is greater than.
- Right: 5000.
- True branch = Senior Approval.
- False branch = Fast Track Approval.

Part C - Configure both approvals

Step 4. True branch: Start and wait for an approval - Senior Approval.

- Approval type: Approve/Reject - First to respond.
- Assigned to: instructor-approved internal account.

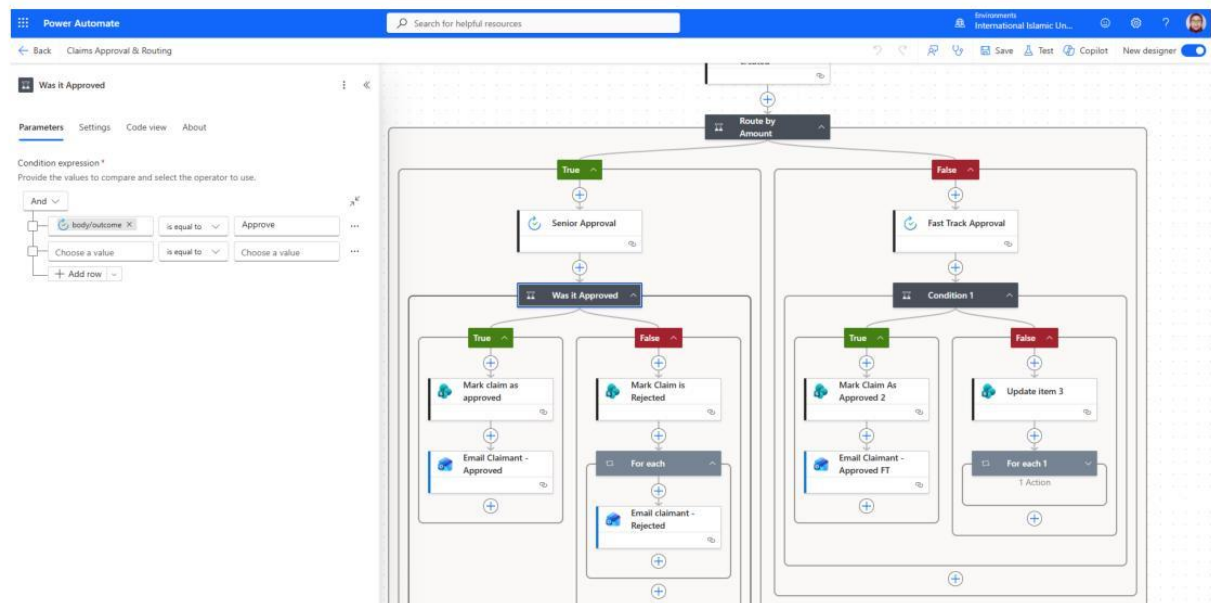
- Title: High-value claim: [ClaimantName] - RM [ClaimAmount].
- Details: Claimant, CertificateNo, ClaimType, IncidentDate, ClaimAmount, Description.
- Item link: Link to item if available.
- Item link description: Open the claim in SharePoint.

Step 5. False branch: Start and wait for an approval - Fast Track Approval.

- Use the same approval type.
- Assigned to: valid fast-track approver/test account.
- Title: Fast-track: [ClaimantName] - RM [ClaimAmount].
- Include the same claim details.

Part D - Handle the approval response

Nested Condition



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Reference slide: nested condition after each approval response.

Step 6. Under Senior Approval, add Condition - Was it approved?

- Left: Outcome from Senior Approval.
- Operator: is equal to.
- Right: Approve.

Step 7. Senior/True: mark claim Approved and notify the claimant.

- SharePoint - Update item: set Status = Approved.
- Outlook - Send an email (V2): subject Your takaful claim has been approved - [CertificateNo].
- Use Dynamic Content for claimant name, certificate number, and amount.

Step 8. Senior/False: mark claim Rejected and notify the claimant.

- SharePoint - Update item: set Status = Rejected.
- Outlook - Send an email (V2): subject Update on your takaful claim - [CertificateNo].
- Include approval comments if appropriate for the training scenario.

Step 9. Repeat Steps 6-8 under Fast Track Approval.

Rename the corresponding condition and actions clearly, for example Fast Track - Was it approved?, Fast Track - Mark approved, and Fast Track - Email rejected.

Part E - Write the audit trail

Step 10. In each final outcome branch, add SharePoint - Create item to Claims_Audit_Log.

- ClaimID: ID from the Takaful_Claims_Intake trigger.
- ClaimantName: claimant dynamic content.
- Approver: use the approver/responder dynamic content when available; otherwise use the configured approver email.
- Outcome: Approved or Rejected for that branch.
- Comments: approval comments/response comments.
- DecisionTimestamp: expression utcNow().
- Rename each action: Log decision to audit.

Step 11. Save and run Flow Checker.

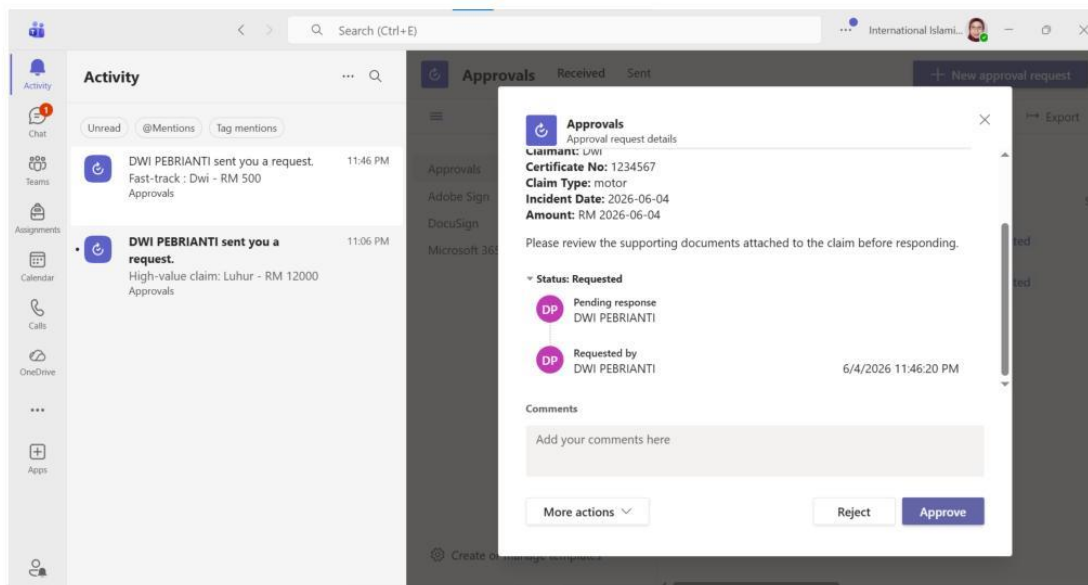
Part F - Test matrix

Test	Amount	Approval path	Human response	Expected final status
A	RM 12,000	Senior	Approve	Approved
B	RM 12,000	Senior	Reject	Rejected
C	RM 500	Fast Track	Approve	Approved
D	RM 500	Fast Track	Reject	Rejected
Edge	RM 5,000	Fast Track (because rule is > 5000)	Either	Matches response

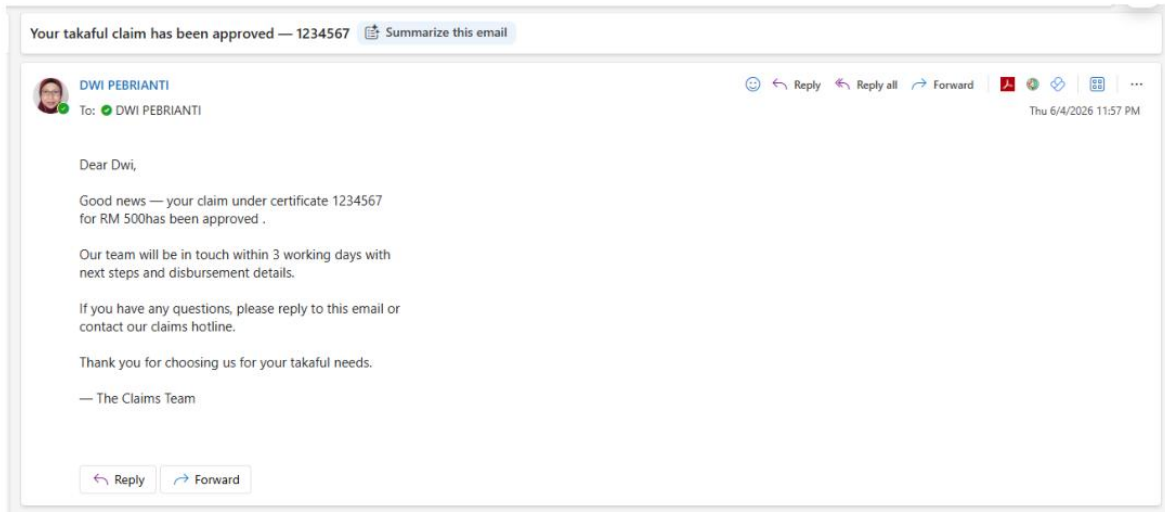
Step 12. For each test, verify four locations.

- Power Automate Run history: trigger, condition, approval, update, email, and audit actions show the expected status.
- Approvals/Teams/Outlook: the approval request displays the correct claim values.
- Takaful_Claims_Intake: Status changes to Approved or Rejected.
- Claims_Audit_Log: one decision record exists with the correct outcome and timestamp.

Approval Message



In Microsoft Outlook



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Reference slide: claimant approval email after the flow completes.

PASS criterion

The flow correctly routes high- and low-value claims, waits for a human decision, updates the intake record, emails the claimant, and writes an audit record. The RM 5,000 boundary behaves exactly as defined by the > 5000 rule.

Expected waiting behavior

A run showing “Running” at Start and wait for an approval is not automatically an error. The flow is intentionally paused until the assigned approver responds.

Lab 10 - Final Validation, Troubleshooting and Submission

The training slides emphasize: never trust a flow because one run succeeded. Validate the design, the happy path, and the edge cases.

A. Universal validation sequence

1. Save the flow.
2. Run Flow Checker and resolve syntax, connection, or missing-parameter errors.
3. Run a manual/controlled test.
4. Open Run history and inspect every action, not only the final email.
5. Confirm external effects: SharePoint row, OneDrive file, email, approval, or audit record.
6. Run at least one negative/False-path case.
7. Run a boundary-value test where a numeric condition is used.
8. After edits, test again to ensure the change did not break earlier behavior.

B. Troubleshooting table

Symptom	Likely cause	Correction
Flow never triggers	Wrong environment, wrong site/list,	Verify environment, connector, Site

	or trigger filter.	Address/List Name, and create a brand-new test item.
Email goes to the wrong person	Recipient was hardcoded.	Use ClaimantEmail Dynamic Content.
Approval fails: no valid users	Assigned to is not a valid internal approver.	Use a valid Microsoft 365 user email/UPN.
Approval flow keeps running	Start and wait for an approval is waiting for a human.	Open Approvals/Outlook and Approve or Reject.
Email shows concat(...) or replace(...)	Expression was typed as ordinary text.	Use fx > Expression and insert the expression token.
HTML table headers are dates/numbers	Select mapping is reversed.	Fixed labels on LEFT; Dynamic Content on RIGHT.
HTML report includes Created/Modified/Author JSON	Create HTML table is using raw Get items output.	Use Select output as the Create HTML table input.
Attachment file is corrupt	Metadata used instead of attachment bytes.	Use Get Attachment (V2) Content Bytes for Create file.
Variable cannot be initialized inside a branch	Initialize variable was placed in a Condition/loop.	Move Initialize variable to top-level immediately after the trigger.

C. Student evidence checklist

- Screenshot of the final flow structure.
- Screenshot of one successful Run history.
- Screenshot of one branch/edge-case test.
- Evidence of the business result: email, saved file, SharePoint update, or approval.
- For the capstone: screenshot of Takaful_Claims_Intake status and the matching Claims_Audit_Log row.
- One sentence explaining what the automation replaced and how you verified it.

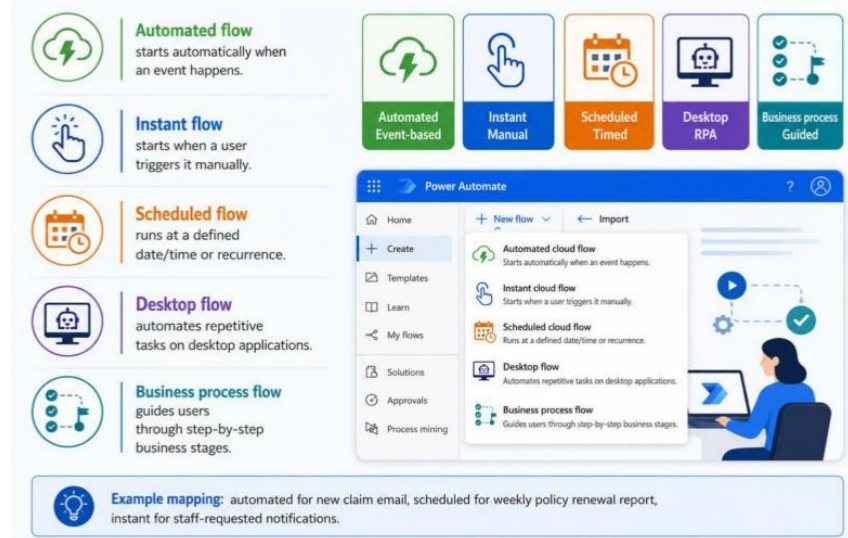
D. Completion rubric

Criterion	Excellent	Acceptable	Needs correction
Flow selection	Correct flow type and trigger match business event.	Flow works but selection is inefficient.	Wrong trigger/flow type.
Configuration	Connections, fields, and Dynamic Content are correct.	Minor non-critical issue.	Hardcoded/incorrect data or broken connection.
Logic	Conditions and branches match exact business rules.	Happy path works; edge case weak.	Incorrect routing.
Validation	Flow Checker + positive + negative + boundary tests.	One successful controlled test.	No reproducible test evidence.
Governance	Clear names, test environment, audit/logging where needed.	Mostly clear naming.	Generic names and no audit thinking.

Appendix A - Quick Flow Selection Guide

MODULE 2 • ECOSYSTEM

Types of Flows



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Reference slide: Automated, Instant, Scheduled, Desktop, and Business Process flows.

Need	Flow type	Trigger example	Training example
Something happens automatically	Automated cloud flow	New SharePoint item / new email	Claim acknowledgement, attachments
A person presses a button	Instant cloud flow	Manually trigger a flow	Generate HTML report / FNOL button
Something runs at a time	Scheduled cloud flow	Recurrence	Daily reminder / weekly summary
Legacy UI has no useful API	Desktop flow	Desktop/manual UI trigger	Legacy insurance portal

Appendix B - Expression Cheat Sheet

Purpose	Expression	Example output
Format date	<code>formatDateTime(triggerBody()?['ClaimDate'],'dd-MMM-yyyy')</code>	15-Jan-2026
Format currency	<code>formatNumber(triggerBody()?['ClaimAmount'],'C2','en-US')</code>	\$5,000.00
Add amount + fee	<code>add(triggerBody()?['ClaimAmount'],triggerBody()?['AdminFee'])</code>	5125
Build claim reference	<code>concat(triggerBody()?['PolicyNumber'],'-',formatDateTime(triggerBody()?['ClaimDate'],'yyyyMMdd'))</code>	POL45821-20260115
Current UTC time	<code>utcNow()</code>	ISO timestamp

Appendix C - Current Microsoft References

The step names and guidance in this manual were checked against Microsoft Learn documentation available on 26 July 2026. The Power Automate interface can vary slightly by tenant, license, and new/classic designer.

- [Overview of cloud flows](#)

- [Run a cloud flow on a schedule](#)
- [Office 365 Outlook connector](#)
- [OneDrive for Business connector](#)
- [SharePoint connector](#)
- [RSS connector](#)
- [Use data operations](#)
- [Store and manage values in variables](#)
- [Create and test an approval workflow](#)
- [Create a cloud flow using Copilot](#)